

Content Outline

At a bare minimum, your 20-25 minute presentation should cover the following:

- A background of the chosen city, unique aspects of that city and its energy needs
- A discussion of the renewables chosen and why
- Covering the following topics:
 - Energy needs calculations (sources, assumptions, methods and results)
 - Renewable energy system modeling (assumptions, methods and results)
 - Energy storage modeling/sizing (assumptions, methods and results)
 - Economic analysis (sources, assumptions, methods, and results)
- Some discussion on the feasibility and implementation of the combination of energy sources
 - Where would this be located, does the geography allow for this, etc.
- Limitations and challenges to the project. Limitations of your study and your methods
- Summary and conclusions.
- **The report should contain similar content and be formatted according to the posted template.**

APRIL 28, 2025



Designing a Net-Zero Electricity Future for Los Angeles: A Technical and Economic Feasibility Study

Sustainable Energy for the Built Environment - CE397

Aritro De, Calvin Lin, Daniel Sung

Background: Los Angeles

- 2nd-largest city in the U.S. (population ~3.8 million in 2023)
- Metro area: ~12.8 million people, GDP ~\$1.3 trillion
- One of the world's largest urban economies
- Sources: U.S. Bureau of Economic Analysis (2023)



The LADWP

- Largest municipal utility in the nation
- ~8,100 MW generation capacity
- ~26 million MWh delivered yearly
- ~1.4–1.5 million customer accounts

Current Emissions Progress

- LA's GHG emissions: ~26.2 million metric tons CO₂e (2022)
- Stationary energy: ~15.6 MMT CO₂e (2022)
- 30% decrease from 1990 levels overall
- Coal-fired capacity retired, renewables on the rise
- 45% renewable electricity in 2023 (up from 33% in 2020)

Drivers for Further Decarbonization

- Air quality (ozone, particulate matter)
- Public health benefits: reduced respiratory diseases
- Meeting/exceeding state mandates (e.g., SB 100 in California)
- Economic opportunity in renewable infrastructure

Project Goal

- Achieve net-zero electricity in Los Angeles
- Focus on **solar** and **wind** additions
- Displace non-renewable generation across a full year
- Build on LADWP's existing load/demand data

Methodology

- Analyzing LA's monthly/hourly demand
- Evaluating solar & wind potential
- Calculating needed capacity
- Conducting economic feasibility

Energy Demand

- LADWP operational data from EIA Open Data platform
- Region-level vs. fuel-type datasets
- Goal: Understand how supply meets demand

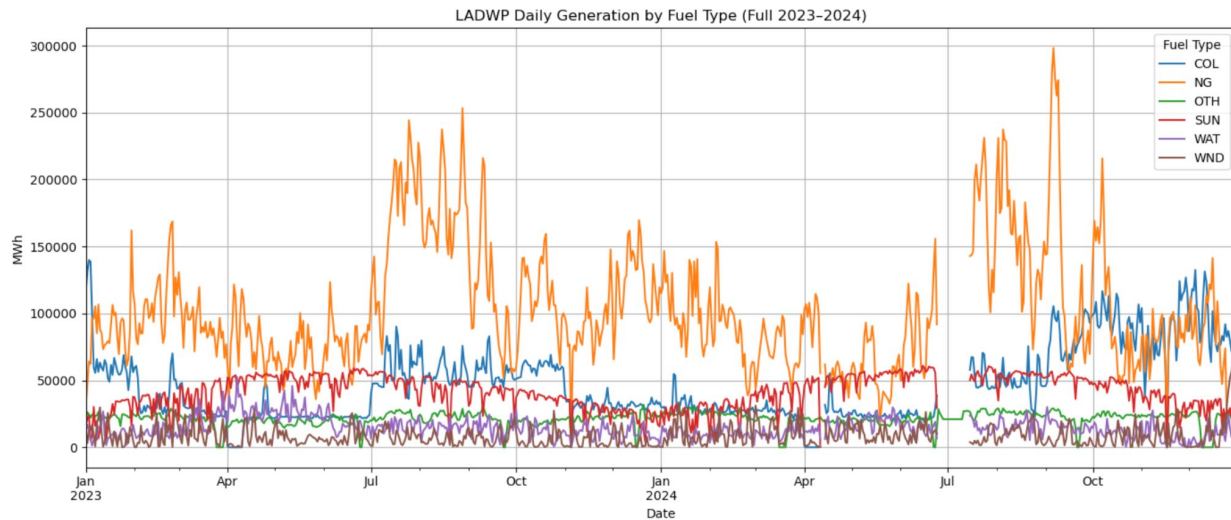
Region-Level Data

- Total Demand (D)
- Forecast Demand (DF)
- Net Generation (NG)
- Interchange (TI) (*not used in our project*)

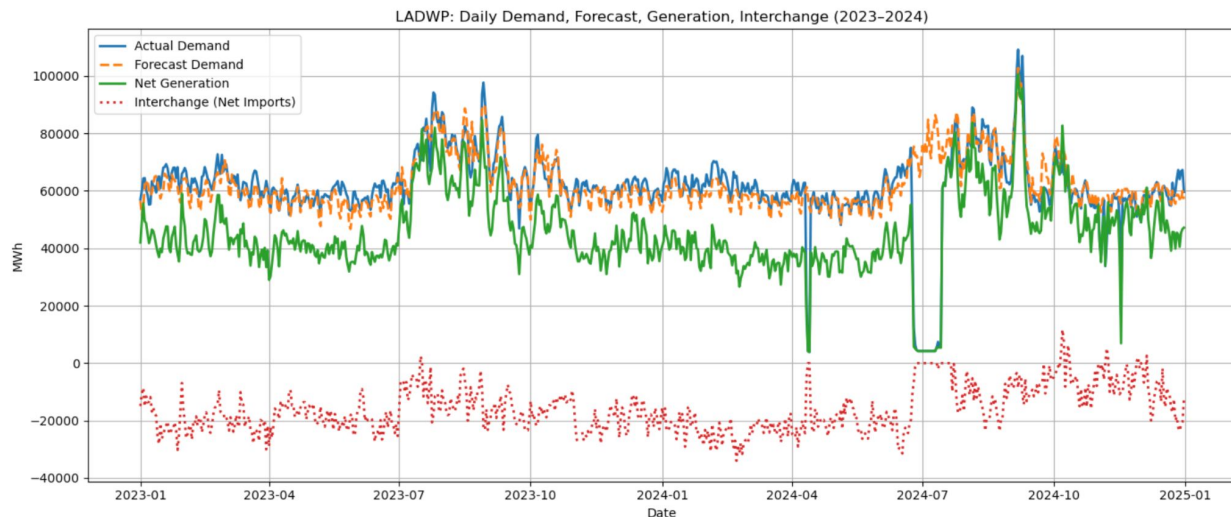
Fuel-Type Data

- Coal (COL), Natural Gas (NG), Solar (SUN), Wind (WND), Hydro (WAT), Other (OTH)

LADWP daily generation by fuel type

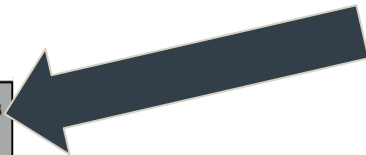


LADWP daily demand, forecast, generation, and interchange



Combined Hourly LADWP Dataset

Date	D	COL	NatGas	OTH	SUN	WAT	WND	diff_demand_ minus_netgen	non_ren ewable	diff_demand_mi nus_nonrenewa ble	renewabl e	diff_demand_minus _renewable
2023-01-01 00:00:00	2712.0	916.0	250.0	212.0	66.0	0.0	255.0	1013.0	1378.0	1334.0	321.0	2391.0
2023-01-01 01:00:00	2813.0	1051. 0	392.0	220.0	10.0	0.0	243.0	897.0	1663.0	1150.0	253.0	2560.0
2023-01-01 02:00:00	2801.0	1069. 0	474.0	223.0	0.0	0.0	234.0	801.0	1766.0	1035.0	234.0	2567.0
2023-01-01 03:00:00	2824.0	1066. 0	381.0	222.0	0.0	0.0	196.0	959.0	1669.0	1155.0	196.0	2628.0
2023-01-01 04:00:00	2782.0	1144. 0	381.0	219.0	0.0	0.0	239.0	799.0	1744.0	1038.0	239.0	2543.0



Selection of Renewables

High Solar Potential

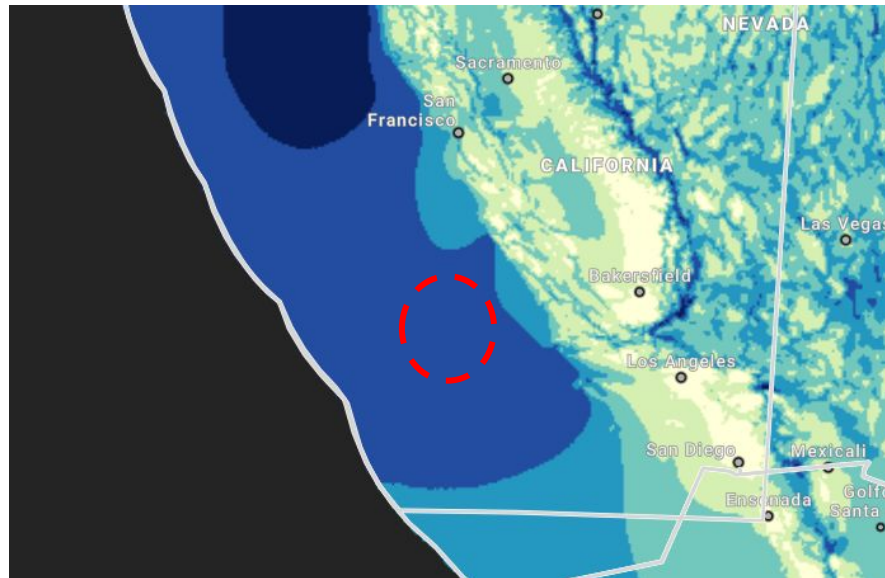
- Sunny climate in Los Angeles
- Consistent irradiance throughout the year

Viable Wind Resources

- **Island site** selected for offshore wind (strong coastal breezes)

Wind

- Data from NREL Wind Resource DB API, data availability - 2019 (latest)
- Site: Offshore LA.
- Co-ordinates: -121.5 34.12



Process - wind speed

- Selected three different height for wind speeds
 - 80 meters
 - 100 meters
 - 120 meters
- Selected optimal height based on max average wind speed monthly.

Wind Turbine Model

- Model: GE Vernova GE 2.5 - 120
- Key parameters
 - rated power: 2.5 MW,
 - cut in : 3.0 m/s,
 - rated : 12.0,
 - cut out : 25.0 m/s,
 - hub height : 80, 110, 124m
 - rotor dia : 120m



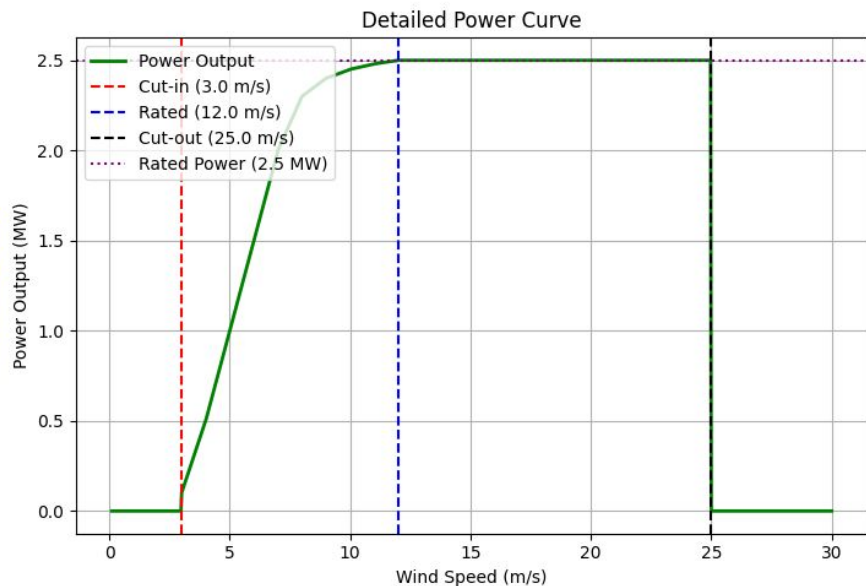
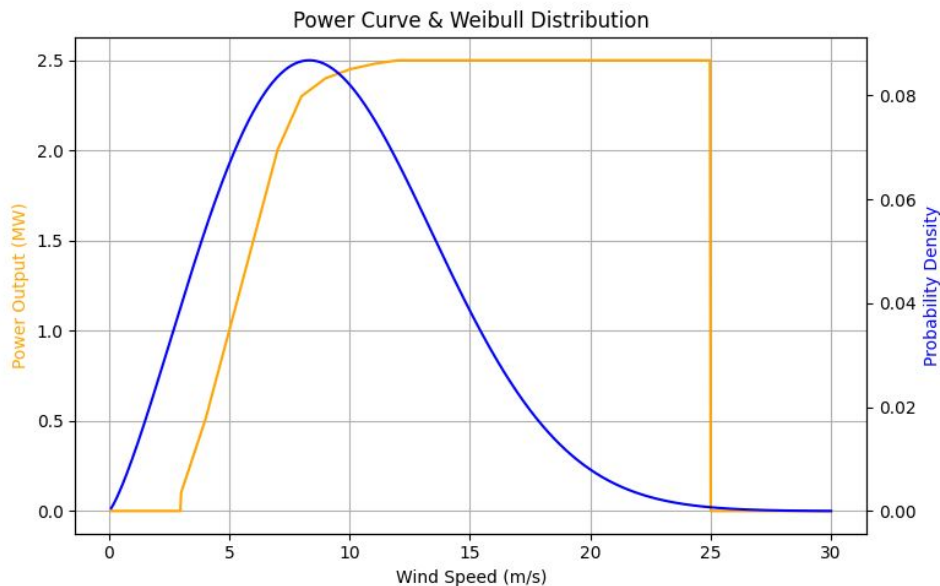
Process - Weibull distribution

$$f(x; k, c) = \frac{k}{c} \left(\frac{x}{c}\right)^{k-1} e^{-(x/c)^k}$$

where $x \geq 0$, $k > 0$ is the shape parameter, and $c > 0$ is the scale parameter.

- Found a function in scipy - weibull_min

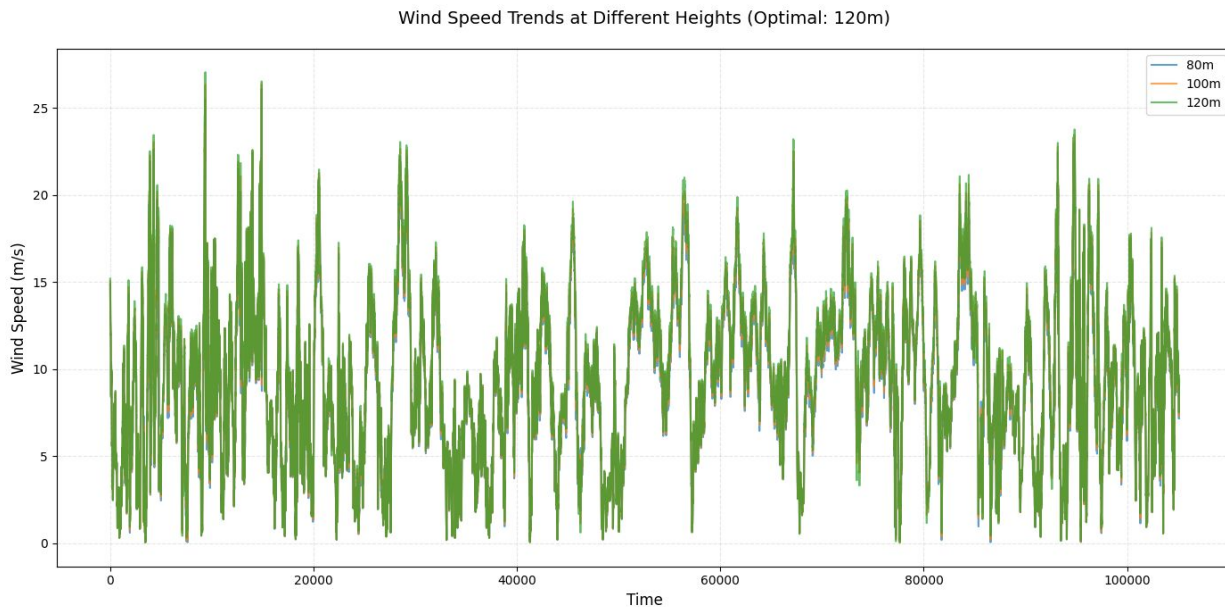
Power curve & Weibull distribution



Selection of turbine - progression

- Currently selected
GE Vernova 2.5MW
- However, to pull data from turbine Database
(<https://en.wind-turbine-models.com/turbines>) for optimal wind turbines based on height, power-curve and rated power

Wind speed different heights

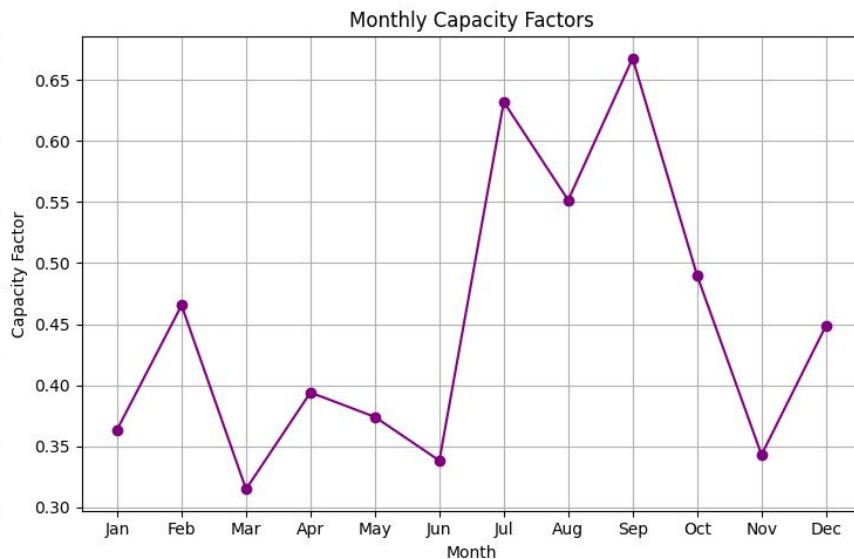
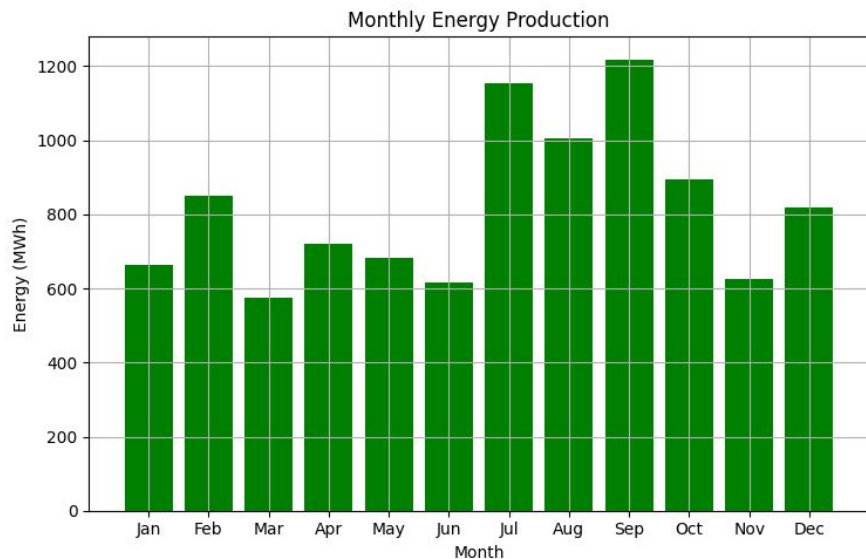


Turbine performance

- Annual Energy Generation
- Capacity Factor of the turbine

Annual Energy
Generation: 9.83 GWh
Capacity factor: 55.2%

Monthly energy & Capacity Factors



- For 997 turbines, 9.80 TWh is generated annually

Solar Data

- National Solar Radiation Database (NSRDB)
- Site: desert location northeast of LA.
- Year of data: 2022

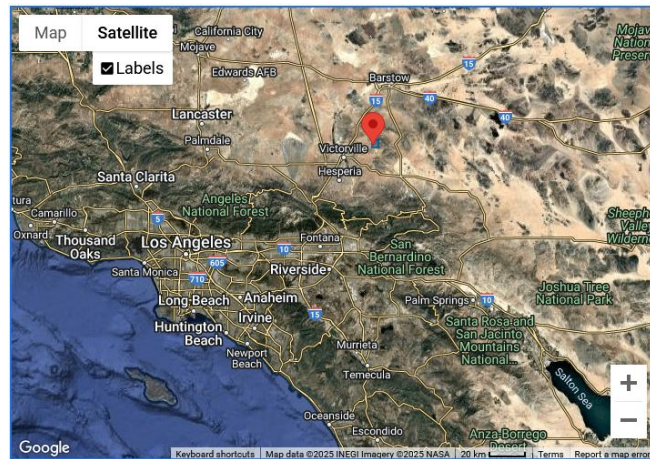
SOLAR RESOURCE DATA

The latitude and longitude of the solar resource data site is shown below, along with the distance between your location and the center of the site grid cell. Use this data unless you have a reason to change it.

Solar resource
data site

Lat, Lng: 34.61, -117.1

1.3 mi



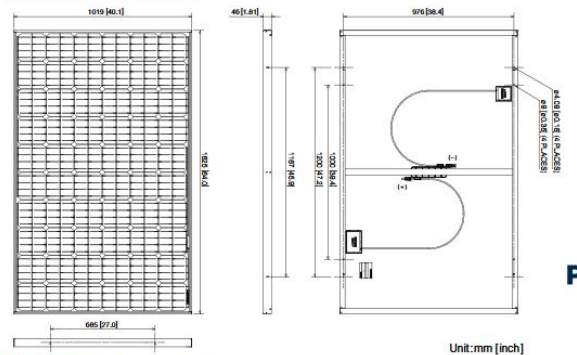
(34.59593337169398, -117.11428489257814)

Images from NREL's PVWatts tool

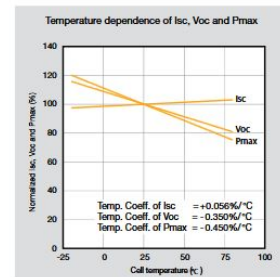
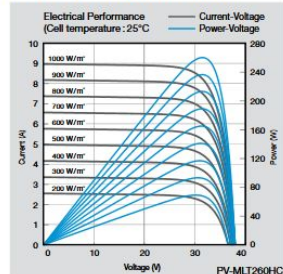
Solar Panel Model

- Model: PV-MLU250HC (Same as class)
- Key parameters
 - Module efficiency: 15.1%
 - Rated power: 250 W
 - Area: 1.66 m²
 - NOCT: 45.7
- Has maximum power point tracker with efficiency of 0.9.

DRAWINGS AND DIMENSIONS



ELECTRICAL CHARACTERISTICS



Solar Calculations

- Same as HW4
- Panel is fixed and south facing.
- Slope is latitude (34.595).

$$(\tau\alpha)G_T = \eta_c G_T + U_L(T_c - T_a)$$

$$(\tau\alpha)G_{NOCT} = U_L(NOCT - 20)$$

$$\eta = \eta_{ref} [1 - \beta(T_c - T_{c,ref})] \eta_{MPPT}$$

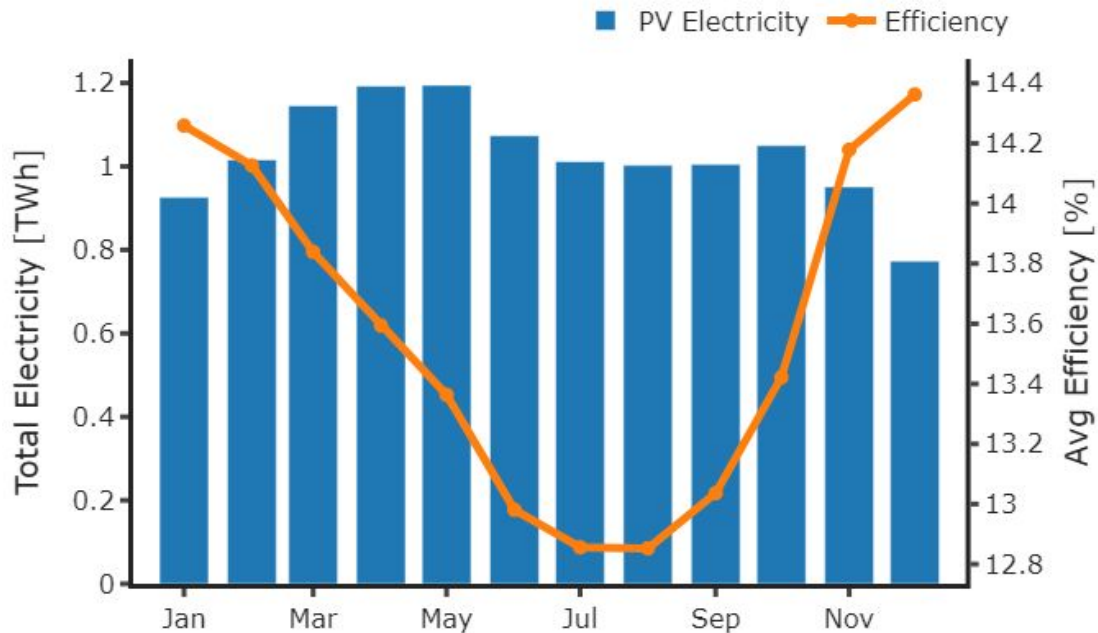
$$\eta = \eta_c * \eta_{MPPT}$$

$$W_E = \eta G_T A$$

Solar Generation

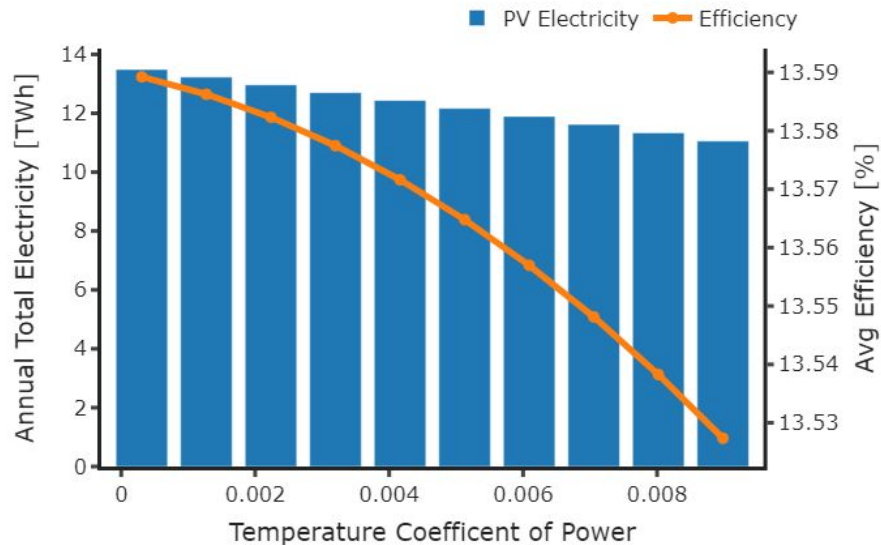
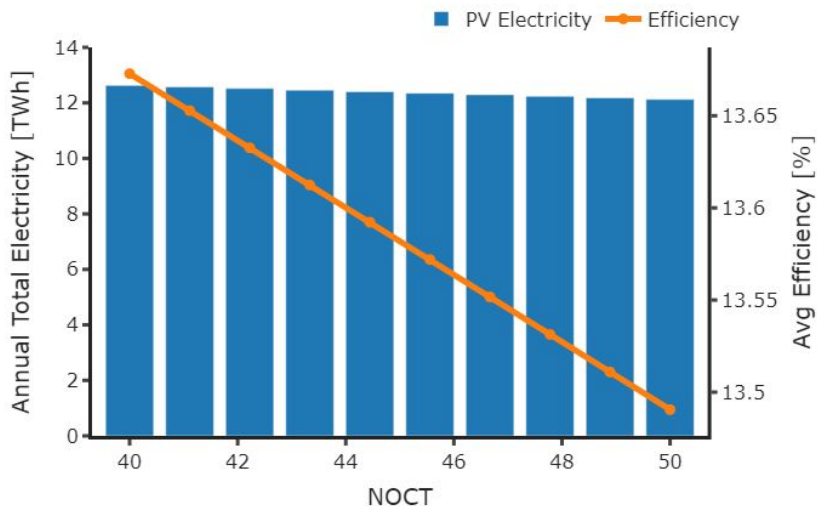
For 24 million panels, 12.3 TWh is generated annually

Solar Generation and Efficiency



Sensitivity Analysis

- Value range from the California Energy Commission's list of PV modules
- Annual generation calculated for each variation. Other parameters unchanged.



Sizing Wind–Solar Solutions for the Peak Monthly Shortfall

- Identified **worst-case** monthly shortfall from dataset
- Ensures covering the most demanding scenario
- Peak shortfall: 2,005,339 MWh in August 2023

Combined Monthly LADWP Dataset



Date	D	COL	NatGas	OTH	SUN	WAT	WND	diff_dem and_min us_netge n	non_renew able	diff_dema nd_minus _nonrene wable	renewable	diff_deman d_minus_r enewable
2023-01-31	1963949.0	432860.0	524552.0	143278.0	148037.0	61200.0	62102.0	591920.0	1100690.0	863259.0	271339.0	1692610.0
2023-02-28	1806346.0	210541.0	613248.0	132492.0	184941.0	59406.0	49454.0	556264.0	956281.0	850065.0	293801.0	1512545.0
2023-03-31	1937027.0	179695.0	571257.0	111080.0	231396.0	124008.0	68924.0	650667.0	862032.0	1074995.0	424328.0	1512699.0
2023-04-30	1726237.0	102747.0	479265.0	119081.0	310892.0	171192.0	40845.0	502215.0	701093.0	1025144.0	522929.0	1203308.0
2023-05-31	1810039.0	141046.0	405117.0	110825.0	320495.0	169138.0	44737.0	618681.0	656988.0	1153051.0	534370.0	1275669.0

50–50 Scenario

- **50% solar / 50% wind case (August)**
- **1 solar set (16 panels) $\Rightarrow$ ~0.71 MWh/month**
- **1 wind turbine $\Rightarrow$ ~1006.48 MWh/month**
- **~1,418,686 solar sets & 997 turbines**

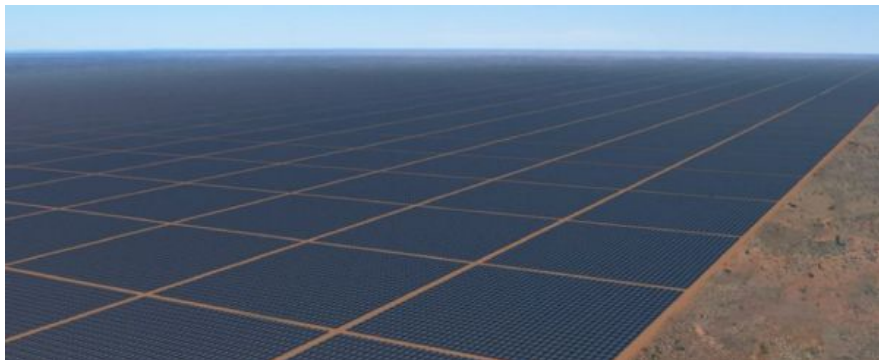
Solar Site/Plant

- 24 million panels need to meet requirements.
 - 39.9 km² area
 - 1.002 TWh gen for Aug.
- Solar data site has space of 6.80 km².
 - Additional sites needed.
- Rated power: 6 GW
- Annual gen of 12.3 TWh.



Is our solar farm feasible?

- Our design is:
 - 24 million panels
 - 6 GW
- The largest solar farm in CA is the Solar Star
 - 1.2 million panels
 - 579 MW
- Current largest in the world is the Xinjiang Midong Solar Park in China at 5 GW.
- Proposed 10 GW farm in Australia at cost of AUD \$20B



A rendering of proposed solar farm in Australia.
(Sun Cable)

Solar Site Cost

- \$7.81 Billion install cost (at \$1.3/W)



- In 2023, the ranges in average U.S. PV system pricing across methods were reported to be:
 - \$2.3/W–\$4.4/W for residential solar
 - \$1.5/W–\$3.2/W for nonresidential solar
 - \$1.0/W–\$1.3/W for utility-scale solar.

Economic Analysis

- Wholesale electric price: \$35.90/MWh (from CAISO)
 - Inflation: 1.5%
- Total System Cost (C_S): \$15.04B
 - Wind farm: \$7.23B
 - Solar farm: \$7.81B
- Total Grid-Equivalent Energy Cost (C_E): \$793 M
 - Solar energy: 9.8 TWh
 - Wind energy: 12.3 TWh
- 20 year loan at 5% interest and 0.25 downpayment
- 25 year analysis period
- 0.04 maintenance ratio at 1.5% inflation
- 7% market discount rate
- No tax rebates

Result

- $LCC_{grid} = \$1.06 \text{ T}$, $LCC_{farm} = \$21.3 \text{ T}$
- Savings of -\$22 B
- Payback period: unachievable, savings get more negative with increased analysis length
- LCOE: \$386.66/MWh
- May need further validation

$$P_1 = (1 - ct)PWF(N, i, d)$$

$$P_2 = D - R + (1 - D) \frac{PWF(N_1, 0, d)}{PWF(N_L, 0, m)} + [M_s(1 - ct)]PWF(N, i, d)$$

$$C_E = grid_{price} \cdot annual \text{ energy}$$

$$C_S = \text{first year coast}$$

$$LCC_{farm} = P_2 C_S$$

$$LCC_{grid} = P_1 C_E$$

$$LCS = LCC_{grid} - LCC_{farm}$$

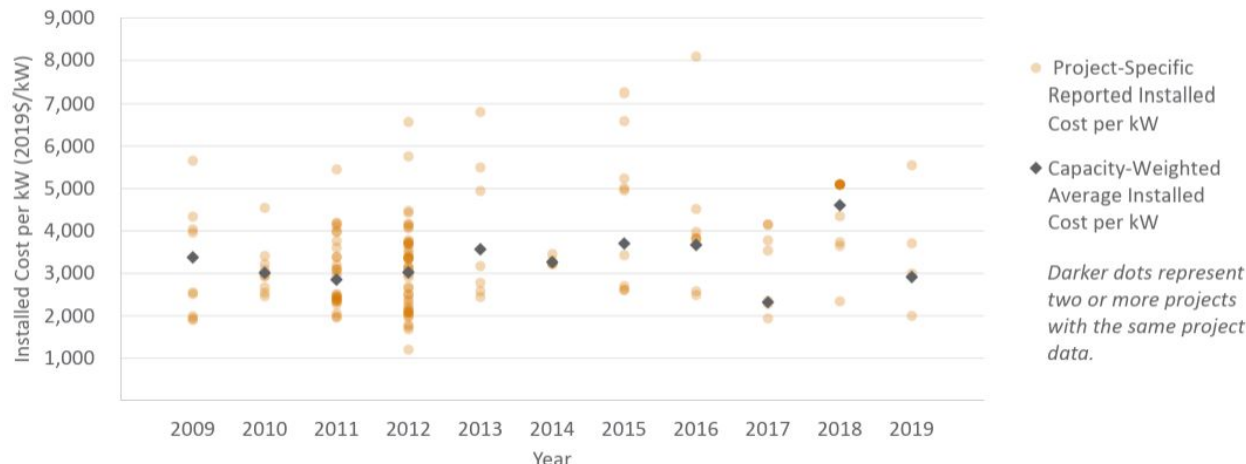
Wind Site Cost

- \$7.23 Billion install cost (at \$2900/kW)
- (2.5 MW x \$2.9M x 997 turbines)



U.S. Distributed Wind Installed Costs: Turbines Greater than 100 kW

- Annual average capacity-weighted installed costs for projects using **turbines greater than 100 kW** range from around \$2,000 to \$5,000 per kW with the 2019 capacity-weighted average cost around \$2,900 per kW. The small sample sizes and high variance in project-specific costs both contribute to this wide cost range, although the range is smaller than it is for small wind.



Summary/Conclusion

Conclusion

- LA can meet its electricity demand from renewables
- Largest shortfall occurs in August (~2 million MWh)
- Solar and wind together offer complementary generation

Feasibility & Scale

- Substantial capacity needed (millions of panels, hundreds of turbines)
- Requires robust infrastructure and grid integration

Economic & Environmental Benefits

- Reduced fossil dependence
- Improved air quality and public health
- Potential for job creation in clean energy

Next Steps

- Evaluate land-use or offshore permitting
- Explore energy storage for peak load balancing

Limitations & Future Improvements

- Energy storage
- Energy transmission
- Find the best combination of solar & wind

Results of Wind–Solar Splits for the Peak Monthly Shortfall

	Solar_%	Wind_%	Num_Solar_Panels	Area_Solar_m2	Num_Turbines
0	0	100	0	0.000000e+00	1993
1	1	99	480784	7.979812e+05	1973
2	2	98	961552	1.595936e+06	1953
3	3	97	1442320	2.393891e+06	1933
4	4	96	1923088	3.191845e+06	1913
...	...	...	...	...	...
96	96	4	46153888	7.660392e+07	80
97	97	3	46634656	7.740187e+07	60
98	98	2	47115424	7.819982e+07	40
99	99	1	47596192	7.899778e+07	20
100	100	0	48076976	7.979576e+07	0